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Simplified Mathematical Representations of Single Shaft Gas Turbines in Mechanical Drive Service

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ABSTRACT

There have been several recent applications of large single shaft, heavy duty gas turbines in mechanical drive service, powering high horsepower, multi-casing compressors. This variable speed application of a traditional constant speed driver, with a more limited operating speed range, has created a need for simplified but accurate mathematical representations that can be incorporated into overall process simulations to allow interactive dynamic evaluation of the complete system.

This paper presents simplified mathematical representations of four gas turbines covering the horsepower range from 26,000 HP to 108,000 HP. The models incorporate both the control and fuel system characteristics as well as those of the turbomachinery. Although gas fuel is assumed, listed references can be used to accommodate liquid fuel. The models are suitable for a wide range of ambient temperatures, and the influence of axial flow compressor variable inlet guide vanes is included in the models as appropriate to the actual machinery configuration.

NDMENCLATURE

FSR = Per Unit Fuel Command
Ligv = Per Unit Inlet Guide Vane Actuator Stroke
IGV = Actual Inlet Guide Vane Angle - Circular Degrees
N = Per Unit Turbine Rotor Speed
s = Laplace Operator
Ta = Site Ambient Temperature - Degrees F or C
Tm = Measured Turbine Exhaust Temperature - Degrees F or C
Tr = ISD Rated Turbine Exhaust Temperature - Degrees F or C
Tra = Rated Exhaust Temperature at Ta - Degrees F or C
Tx = Actual Turbine Exhaust Temperature - Degrees F or C
Wf = Per Unit Fuel Flow
Wx = Per Unit Exhaust Flow
Ecr = Combustion Reaction Time Delay - seconds
Etd = Turbine & Exhaust system Transport Delay - seconds
Tcd = Compressor Discharge Volume Time Constant - seconds
Ti = Turbine & Load Compressor Rotor Time Constant - seconds
Tt = Temperature Limiter Integration Rate - seconds

INTRODUCTION

The recent utilization of high horsepower process compressors in locations where steam turbine drivers are not practical has resulted in a renewed interest in heavy duty, single shaft gas turbines as drivers for this equipment. Conventional multi-shaft gas turbines generally do not attain the required output ratings. However, the single shaft gas turbine has been in use as a generator drive machine in utility and industrial power generation service for many years, and is now available as a proven prime mover in output ratings that easily exceed the requirements of today's process compressors. Current applications include driving multi-casing compressors for oil field gas re-injection service, and refrigeration compressor drives for LNG liquifaction service.

Single shaft gas turbines are inherently simpler than multi-shaft turbines, since there are fewer and more accessible bearings, and no variable turbine geometry in the gas turbine hot section. These constructional advantages are somewhat offset by the operational disadvantage of a significantly narrower operating speed range, the latter requiring unique process controls to meet the load compressor operating range. In addition, the fact that the gas turbine must startup with the load compressor connected results in both a substantial increase in gas turbine starting power and special process compressor valving and controls to minimize the connected starting load. Nevertheless, the single shaft gas turbine is the only means of achieving the very high output power levels in a "dry" cycle, since the largest heavy duty multi-shaft gas turbines are rated at less than 40,000 HP.

Prior work on simplified simulations for this type of gas turbine (Rowen, 1983), concentrated on the generator drive applications where operation was limited to a narrow range of turbine speed. In addition, the earlier work assumed both a constant compressor inlet guide vane angle and a constant ambient temperature. This paper circumvents these limitations in the context of the more general case of the variable speed, mechanical drive application.

SINGLE SHAFT MECHANICAL DRIVE TURBINE CHARACTERISTICS

The design point turbine characteristics for the single shaft mechanical drive product line are shown in Table I in both english and metric units as a function of Model Series. The information shown is for natural gas fuel, since it is the most common fuel utilized in mechanical drive applications. These characteristics include both fuel flow and exhaust flow, and together with speed, power and torque, represent the "1.0 Per Unit" values utilized throughout the balance of this paper. This type of notation is common in control system analysis, and is really the decimal equivalent of "100 Per Cent". For mechanical drive applications, these turbines are rated at operating speeds of 95% of the corresponding generator drive speeds shown in Rowen, (1983) in order to allow continuous operation at 105% of the mechanical drive rated speed without exceeding the 100% rating of the generator drive application. Thus a 5100 RPM rated generator drive turbine is rated at approximately 4860 RPM for mechanical drive service.

The rotor inertias shown are for the turbine only, and must have the load equipment inertias, referenced to turbine speed, added to the value shown in order to obtain the complete rotating inertia. In the absence of known driven equipment characteristics, the author's "rule of thumb" is that the typical load inertia will equal about 40% of the turbine inertia, but will be higher if gear is present in the system.

Figure 1 shows a single shaft gas turbine of the MS6001B type, nominally rated at 50,000 HP. All four gas turbine series that are available for mechanical drive service are similar in arrangement, and differ only in physical size and the operating characteristics shown in Table I. Atmospheric air enters the compressor inlet shown at point(A) from the inlet filter system. The air then passes through the compressor inlet guide vanes shown at point(B). The air enters the fifteen to seventeen stage gas turbine axial flow compressor where it is compressed by about ten to one, and discharged into the multiple combustion chambers, indicated by point(C).

Fuel is injected and burned, raising the air temperature to that required to meet the output power requirements, up to the design firing temperature limit. The heated, high pressure air is

TABLE 1. TURBINE CHARACTERISTICS AND CONSTANTS*

English Units

Model Series	Turbine Speed	Output Power	Fuel Flow	Exhaust Temp.	Exhaust Flow	Output Torque	Rotor Inertia
	RPM	HP	Btu/sec	Deg F	Lb/sec	Fi-Lb	Lb-Fi ²
5001R	4860	26,400	68,790	988	205	27,200	14,830
5001P	4860	34,000	86,650	925	253	35,000	21,770
6001B	4860	50,000	109,630	1022	284	51,500	27,540
7001EA	3460	108,200	235,330	1001	615	157,800	145,900

Metric Units

	RPM	KW	KJ/sec	Deg C	Kg/sec	Kg-M	Kg-M ²
5001R	4860	19,686	72,580	531	93	3,760	626
5001P	4860	25,354	91,430	496	115	4,839	919
6001B	4860	37,285	115,670	550	129	7,120	1,162
7001EA	3460	80,685	248,310	538	279	21,815	6,157

1. Turbine constants are subject to change without notice
2. Fuel flows shown correspond to indicated power outputs.
3. Rotor inertia shown is for gas turbine only. Driven equipment inertia must be added to obtain total.
4. ISO Rated values are shown for natural gas fuel. Distillate ratings are 2% lower.

then expanded through either a two or three stage gas turbine expander, indicated at point(D), producing sufficient mechanical power to drive both the axial flow compressor and the connected load, all of which are connected and act as a single rotating element. Finally, the exhaust gases pass through the turbine diffuser and turning vanes, past the exhaust thermocouples, shown at point(E) and then to the atmosphere for simple cycle turbines, or the heat recovery equipment for combined cycle systems.

The fact that all rotating elements are coupled together and rotate in unison has a significant effect on the off design point and part load operating characteristics of the system. First of all, the gas turbine must be started with the load connected, which requires that the starting equipment be sized sufficiently large enough to accommodate the additional torque imposed by the driven equipment until the gas turbine reaches self-sustaining conditions. In most cases, special provisions are made in the driven equipment control valving to reduce this added drag to a minimum consistent with reasonable capital costs. Both gas turbine starting and speed range limitations are discussed in more detail in Ekstrom and Garrison, (1991). The operating speed range of the single shaft, mechanical drive gas turbine is limited for two reasons. Both are related to the fact that as the gas turbine slows down under load, the axial flow compressor also slows down. As the axial flow compressor slows down, the cycle air flow drops off, reducing the turbine output and cycle pressure ratio. If the turbine is being fired at its maximum design temperature limit, output drops faster than the load, and the turbine cannot recover speed without a reduction in the power requirement of the driven equipment. If the axial flow compressor speed drops below about 90% of rated speed, it enters a speed range where it may be sensitive to compressor surge unless the inlet guide vanes are closed, thus further reducing air flow and output power capability of the gas turbine.

In short, utilization of single shaft gas turbines for variable speed mechanical drive service requires careful coordination of prime mover, driven equipment and connected process valving and controls. This paper is offered with the intent of providing the prime mover information necessary to allow conduct of the dynamic system studies required to ensure that this coordination is achieved.

GAS TURBINE SIMULATION BLOCK DIAGRAM

A generic mathematical representation of the single shaft gas turbine is shown in Figure 2. It is similar to that used in Rowen (1983), but is tailored to the mechanical drive operation. Specific added features of this model are as follows:

1. Only an isochronous governor is shown, since this is the preferred configuration for mechanical drive turbines.
2. Since natural gas is the preferred fuel, only a single gas fuel system has been modeled.
3. A rudimentary load compressor model is included.
4. Turbine exhaust flow is calculated.
5. The actual and rated turbine exhaust temperature, and exhaust flow calculations incorporate an ambient temperature correction.
6. An add-on patch has been included to accommodate the effects of modulating inlet guide vanes.

These features are described in more detail in the following discussion. Notes 1 through 13 specify details applicable to the four model series covered. The speed governor controls

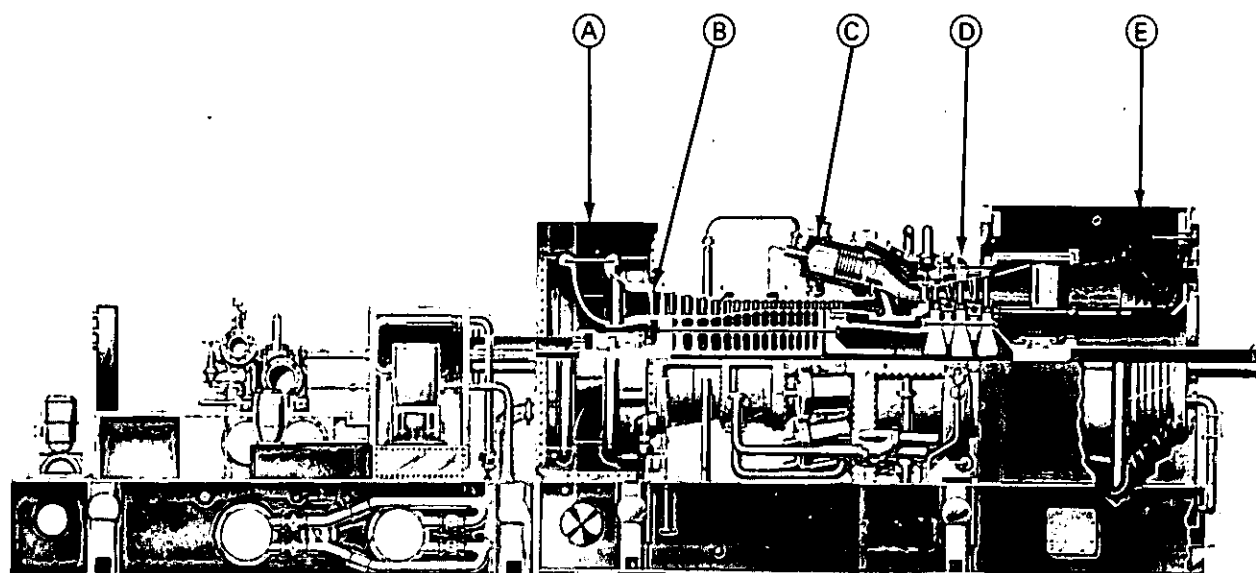


FIG.1. MS6001B SINGLE-SHAFT GAS TURBINE

train speed over the applicable operating range indicated in note 1. This is not the primary means of controlling output, since independent control is required at the load compressor, and is usually achieved by suction throttling, compressor variable geometry, or compressor recirculation. The cumulative effect is represented in the model by the load torque coefficient included in the load compressor simulation.

Control System Representation

The control system includes speed control, temperature control, axial flow compressor inlet guide vane control, acceleration control, and upper and lower fuel limits. The speed Governor representation is for an isochronous speed control, and operates on the speed error formed between a reference made up of one per unit speed plus the digital set point, compared with actual rotor speed. The isochronous governor is a proportional-plus-reset speed controller in which the rate of change of output is proportional to the speed error. Thus, the output of the isochronous governor will integrate in a corrective direction until the speed error is zero.

Exhaust temperature is the normal means of limiting gas turbine output at a predetermined firing temperature, independent of variations in other operating parameters. Part load optimization of exhaust temperature is the primary function of the modulating inlet guide vane control. For heat recovery applications, optimum part load cycle performance is obtained when the turbine exhaust temperature is at a maximum. This is achieved by closing the inlet guide vanes, within the operating range indicated in note 10, at part loads, until the exhaust temperature is at its maximum allowable level.

A bias is provided at partially closed inlet guide vane angles to allow transient over-firing, of up to 30°F in turbine exhaust temperature. This permits the addition of fuel at part load, causing the inlet guide vanes to open, increasing cycle air flow, and ultimately resulting in the turbine output increasing to meet the demands of the speed governor. The degree of momentary overfiring decreases as the inlet guide vanes approach their fully open position, and the gas turbine approaches its capability for existing site and machine conditions. The overfiring has no impact on hot gas path thermal fatigue loading or parts lives since it is of relatively small magnitude, and is quite short in time duration.

For simple cycle turbines, the inlet guide vanes are fully opened at the conclusion of startup, and are essentially fixed over the normal operating range. For these circumstances, the portion of the simulation shown above the $\approx$ symbol can be deleted. See note 13 for additional requirements.

Acceleration control is used primarily during gas turbine startup to limit the rate of rotor acceleration prior to reaching minimum governor speed, thus ameliorating the thermal stresses encountered during startup. In the unlikely event of a rapid unloading of the driven compressor, this control would also act to reduce fuel flow to limit the rate of rise in speed. Minimum and maximum fuel flow limits are established to ensure that fuel flow is not reduced below the point of self-sustaining combustion, nor above that corresponding to the maximum turbine output at site minimum ambient temperature.

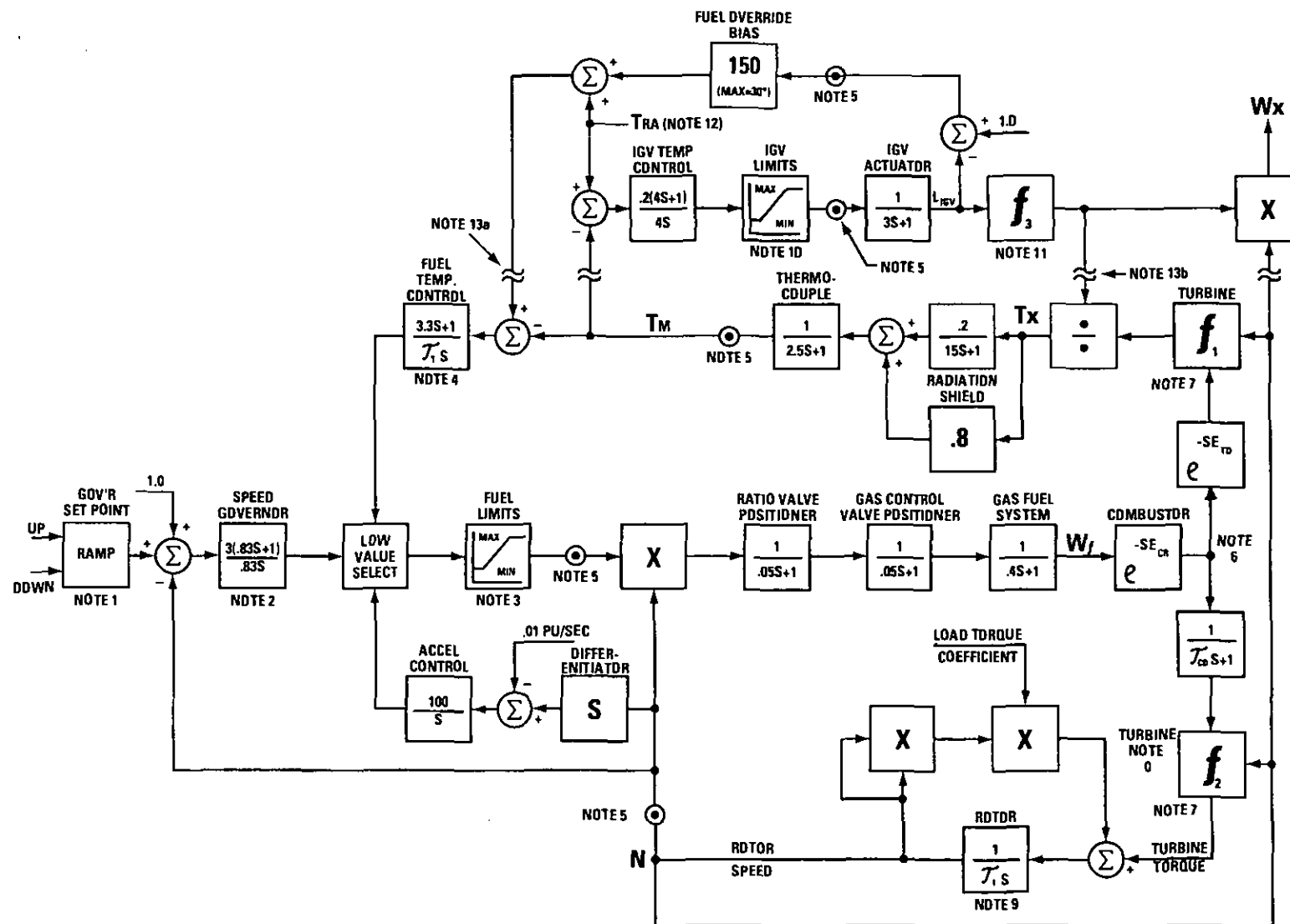


FIG. 2 SIMPLIFIED MATHEMATICAL REPRESENTATION OF SINGLE-SHAFT MECHANICAL DRIVE GAS TURBINE

NOTES FOR FIGURE 2

- Digital setpoint change from low speed stop to high speed stop occurs at a rate of 4% of rated speed per minute. Speed range of operation is as shown below:

Model	Low Speed Stop Setting		High Speed Stop Setting	
	RPM	Per Cent	RPM	Per Cent
5001R	4130	85 %	5105	105 %
5001P	4620	95 %	5105	105 %
6001B	4750	98 %	5105	105 %
7001EA	3420	99 %	3636	105 %

- Speed governor transfer function shown is isochronous.
- Fuel limits are typically Max = +1.0 PU; Min = +0.15 PU.
(No load = +0.2 PU; Rated load = +0.8 PU)
- T₁=450° in Degrees F or 250° in Degrees C.
- For high accuracy applications, five locations designated by the symbol ⊙ should incorporate transport delays of 0.125 seconds for Mark VI, .0625 seconds for Mark V and none for earlier controls.
- Gas turbine dynamic characteristics:

Model	€cr	Tcd	€td
5001R	.01	.10	.02
5001P	.01	.10	.02
6001B	.01	.10	.02
7001EA	.01	.20	.04

- Turbine exhaust temperature calculation

$$f_1 = T_x = \left(\frac{1}{1 + .0027(59 - T_a)} \right) \cdot \left[(T_r - 815(N^2 - 4.21N + 4.42) \cdot .82(1 - Wf)) + 1300(1 - N) + 3.5(\text{MaxIGV} - \text{IGV}) \right] \text{ in } ^\circ\text{F}$$

$$f_1 = T_x = \left(\frac{1}{1 + .0050(15 - T_a)} \right) \cdot \left[(T_r - 453(N^2 - 4.21N + 4.42) \cdot .82(1 - Wf)) + 722(1 - N) + 1.94(\text{MaxIGV} - \text{IGV}) \right] \text{ in } ^\circ\text{C}$$

See note 10 for MaxIGV and IGV

- Turbine torque calculation

$$f_2 = \text{Turbine torque} = (1.16[Wf - .133])/N$$

- Rotating train inertia must include turbine rotor, shown in Table I - Turbine Characteristics, plus driven equipment inertia referenced to turbine rated RPM if a gear is present in the system. Author's rule of thumb is that driven equipment inertia equals 40% of turbine inertia referenced to turbine speed (higher if gear is present.)

- Inlet guide vane limits

Model	MinIGV		MaxIGV	
	Ligv	IGV	Ligv	IGV
5001R	0.00	50	1.00	82
5001P	.33	56	1.00	85
6001B	.46	57	1.00	84
7001EA	.46	57	1.00	84

- Turbine exhaust flow calculation

$$f_3 = W_x = N \cdot [519 / (T_a + 460)] \cdot (\text{Ligv})^{-.257}$$

- The exhaust temperature control point for constant firing temperature is a function of compressor discharge pressure, which is a function of many factors. For these models, this can be expressed purely as a function of ambient temperature as shown below:

$$T_{ra} = T_r - .6(59 - T_a) \text{ in degrees F}$$

$$T_{ra} = T_r - .6(15 - T_a) \text{ in degrees C}$$

Where T_r is obtained from Table I - Turbine Characteristics

- The simple cycle applications of these turbines do not require IGV modulation except during startup, below the minimum operating speed. These situations do not require the variable IGV representation shown above the symbol, but the user must insert T_{ra} at the point designated 13a, and insert 1.0 at the point designated 13b.

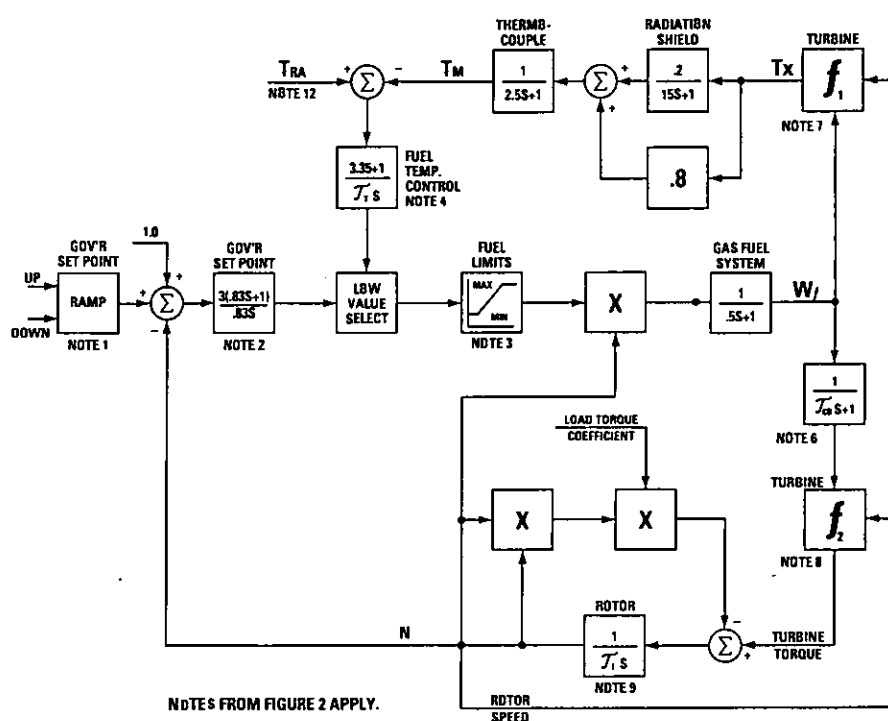


FIG. 3. SIMPLER MATHEMATICAL REPRESENTATION OF SIMPLE CYCLE SINGLE SHAFT MECHANICAL DRIVE GAS TURBINE

Gas Fuel Control System Representation

The fuel gas control system consists of two valves in series, the first of which controls the pressure between the two valves as a function of turbine speed, and is used to extend the controllable turn-down ratio of the system during gas turbine startup. The second valve, the actual gas control valve, has a linear area versus lift characteristic and is aerodynamically designed so that sonic velocities are attained at the controlling area with flange-to-flange pressure ratios as low as 1.25. Since position of the second valve is held proportional to control system fuel command, the fuel flow rate is proportional to the product of fuel command and turbine speed.

The response of the pressure control portion of the system is dominated by the response of the first valve positioner since the volume between the valves is very small, and the pressure measurement transducer has a very small time constant. The gas control valve utilizes a positioner that is identical to the ratio valve, and therefore can be represented by an identical first order transfer function. The remaining time constant in the fuel system is representative of the storage volume of the fuel gas piping between the gas control valve and the fuel nozzles at the gas turbine combustors, and is actually the largest time constant in the gas fuel control system.

Gas Turbine Representation

The gas turbine proper is essentially a linear, non-dynamic device over the specified operating range, with the exception of the rotor inertial time constant. There is a small transport delay associated with the combustion reaction time, a first order time constant associated with the axial flow compressor discharge volume, and a transport delay that is representative of the time required for the combustion gases to move from the combustors through the turbine to the location of the exhaust temperature measurement thermocouples. Both the torque

and exhaust temperature characteristics of the gas turbine are essentially linear with respect to fuel flow, turbine speed and ambient temperature. The torque and exhaust temperature equations are accurate to within .05 per unit at part load, and are significantly more accurate at or near the 1.0 per unit design point.

For applications which utilize modulating inlet guide vanes to optimize part load heat recovery, the effect on turbine performance at constant fuel flow appears only in the exhaust temperature and flow. Turbine output torque is not appreciably effected by guide vane action. Closing the inlet guide vanes at constant part load fuel flow decreases cycle air flow, increases fuel air ratio in the combustor, and causes the exhaust temperature to increase approximately in inverse proportion to the air flow change. The turbine exhaust flow calculation is shown in note 11. The exhaust flow is proportional to a slightly non-linear function of inlet guide vane position, multiplied by the rotor speed, and corrected for ambient temperature. Exhaust flow is provided as an output from the model for use in heat recovery system simulations.

SIMPLIFICATIONS TO THE MODEL

There are several steps that can be taken to simplify the model of Figure 2 with relatively little loss in dynamic accuracy. Simplifications are based on the premise that neither the speed governor set point nor the load torque will change rapidly. This is a reasonable assumption in most cases, since conventional process valving and compressor controls use velocity limited actuators that typically have full travel stroking times of several seconds. The exception is systems that utilize hydraulically actuated variable load compressor geometry, however such systems are not very common. For these stated conditions, the following deletions should be considered:

1. Delete the acceleration control loop, since mechanical drive system transients rarely are fast enough to encounter the acceleration limit. (Load compressor surge is an exception.)
2. Delete the transport delays identified in note 5, since these are too short in duration, relative to the rest of the simulation, to adversely affect the overall dynamic accuracy
3. Assume that the entire fuel gas control system can be represented by a 0.5 second time constant. Again the justification for this simplification arises from the relative comparison of time constant magnitudes.
4. Delete the combustor and exhaust system transport delays for the same reasons as stated above.
5. For simple cycle turbines follow the guide lines of note 13, since the gas turbine axial flow compressor inlet guide vanes do not modulate except during startup, which is beyond the scope of this paper.
6. Conduct the simulation at ISO conditions, since this assumption will allow elimination of the complexities of the ambient temperature correction for exhaust flow and rated and actual exhaust temperatures.

The net result is the simple cycle simple simulation shown in Figure 3. The reader is advised to use discretion in assessing how far to carry the process of simplification. There are no guidelines that can be put forward, since each situation must be evaluated based on the configuration and characteristics of the driven compressor and its connected process equipment.

SUMMARY

This paper discusses many of the features and characteristics that effect the application of heavy duty single shaft gas turbines to variable speed mechanical drive service. It also

offers a highly flexible and fairly accurate, yet still simple mathematical representation of the gas turbine and its fuel and control systems for use in more extensive system simulations of driven equipment and connected processes. Recommended steps for even further simplification are also included.

It is immediately apparent that this simulation is equally applicable to constant speed generator drive service, since this is a special case of the general case described in this paper. This paper offers features, such as modulating inlet guide vanes, calculation of exhaust flow and accomodation of variable ambient temperature, that were not included in Rowen, (1983), which was specific to generator drive service. However, the reader is cautioned that the turbine characteristics of table I are specific to mechanical drive service, as are the isochronous governor, gas fuel system and load equipment representations. Nevertheless, judicious use of the information presented in this paper, in combination with that in Rowen, (1983), will allow a user to simulate any heavy duty single shaft gas turbine in any of the current or envisioned applications.

ACKNOWLEDGEMENT

The author wishes to acknowledge the efforts C. S. McGill, a mechanical engineering student from The Rochester Institute of Technology on co-op assignment with the author's company. Mr. McGill performed most of the data reduction required to develop the thermodynamic characterizations of the gas turbines presented in this paper.

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